

A map can be reachable and still be tactically impossible to read.

Level geometry controls *when* threats become visible. The player's single crosshair has to rotate and service each threat before its combat deadline expires.

See why counting sightlines fails ↓

Geometry → Schedule Pipeline ↓

Map Knowledge Slider (ℓ^*) ↓

Tactical Readability Atlas ↓

📄 Export / Print PDF

Visual 1: Two Rooms — Counting Sightlines Predicts the Wrong Answer Twice

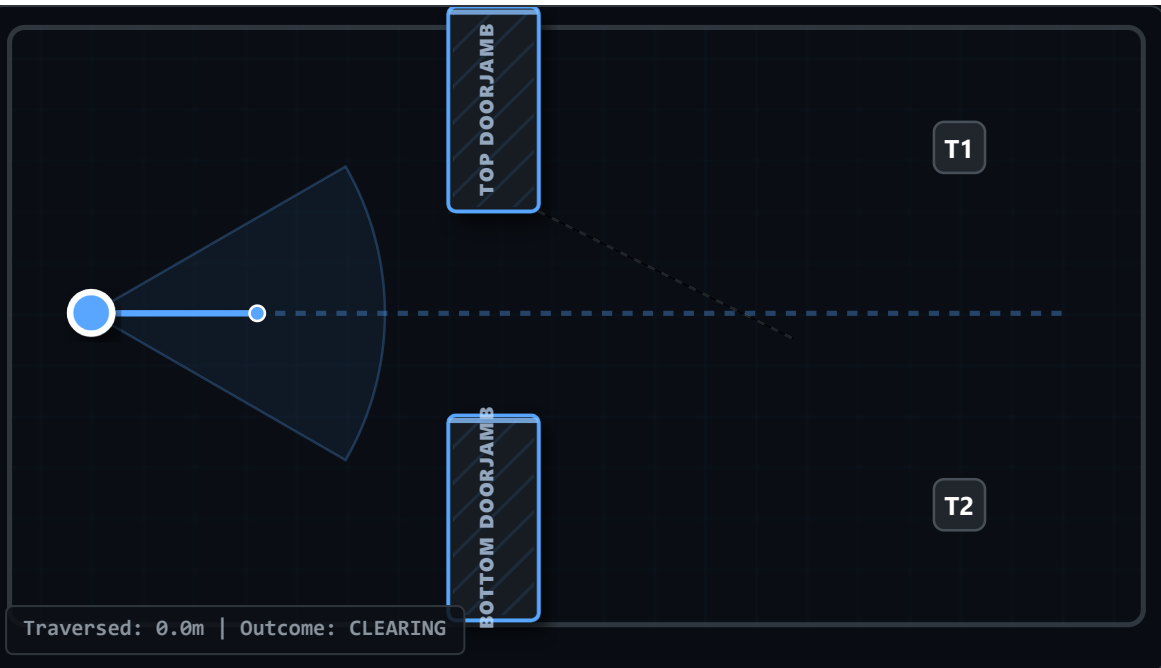
Comparing two layouts where static line-of-sight count (K_{static}) contradicts true temporal clearability.

▶ Play Simulation

↺ Reset

ROOM A · LOW COUNT, UNSERVICEABLE

Only two threats — but the fight is impossible.



SERVICE TIMELINE & DEADLINES

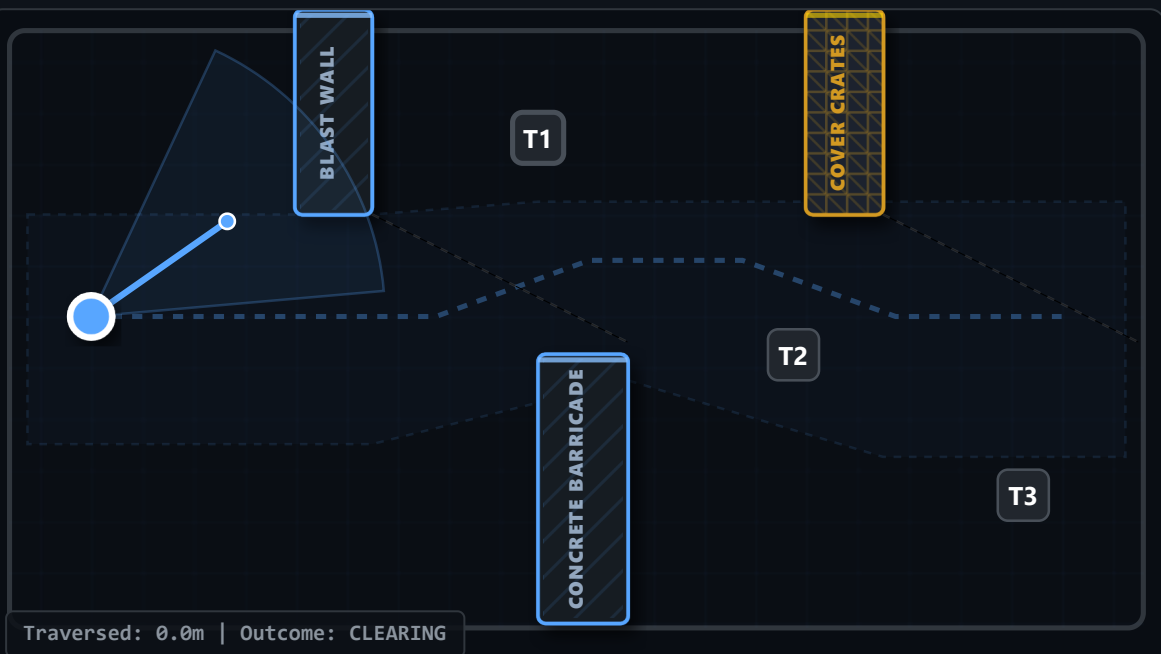


VISIBLE THREATS	STATIC HEURISTIC	TACTICAL MARGIN (M)	OUTCOME
2	PASS ($K \leq 2$)	-1 tic (-28 ms)	UNSERVICEABLE

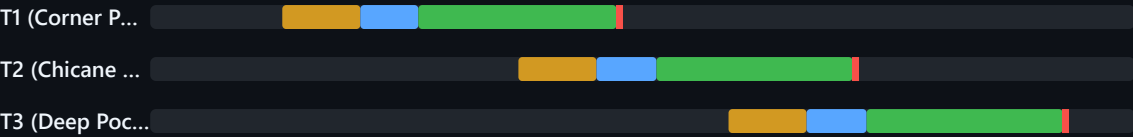
The room does not fail because two targets are visible. It fails because they become urgent at nearly the same time and require a massive 150° reticle flip (417 ms) to service sequentially.

ROOM B · HIGHER COUNT, SERVICEABLE

Three threats — but the fight is clearable.



SERVICE TIMELINE & DEADLINES



VISIBLE THREATS

3

STATIC HEURISTIC

FAIL (K > 2)

TACTICAL MARGIN (M)

+23 tics (+657 ms)

OUTCOME

CLEARABLE ✓

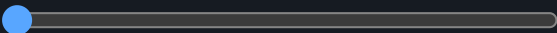
Room B reveals 3 threats sequentially as the player advances. Reticle bearings form a smooth 35° sweep rather than an abrupt cross-screen flip, allowing safe angle isolation.

Counting sightlines predicts the wrong answer twice. Timing and angular sequence matter.

Visual 2: Geometry Creates a Scheduling Problem (The 4-Panel Pipeline)

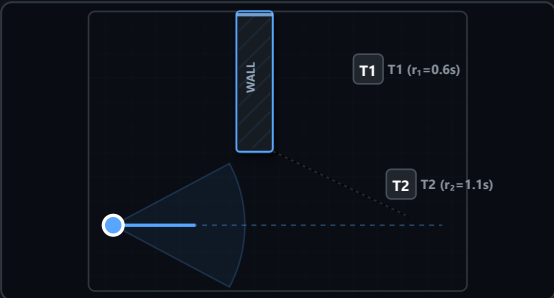
How spatial polygon layout translates into time-sensitive computational jobs on a stateful single-reticle processor.

Time: 0.00s



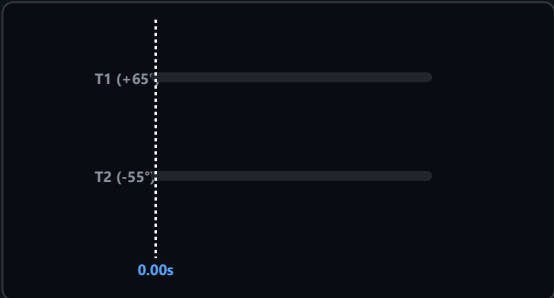
► Play

1 Geometry



1. Geometry decides when each threat becomes visible. Obstacle edges cast grazing rays that trigger visual unocclusion (r_j).

2 Reveal Timeline

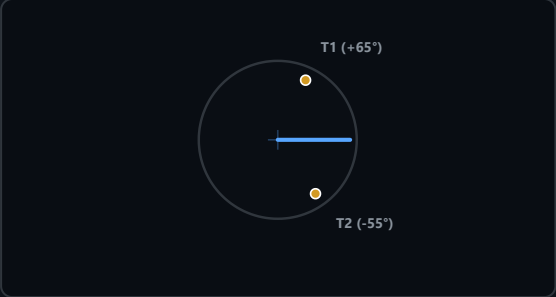


2. Every reveal creates a time-sensitive job. Hostile reaction speed imposes an absolute survival deadline ($D_j = r_j + \eta_j$).

3 Single Reticle

4 Tactical Contract

Required service: 620 ms

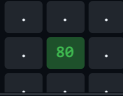


3. One crosshair must service those jobs. Turning farther takes longer ($q_{ij} = \Delta\theta / \omega_{aim}$); next target depends on previous position.

Hostile deadline: 700 ms

Tactical Margin (M): +80 ms ✓

8x8 Composable Transfer Matrix (C_M)

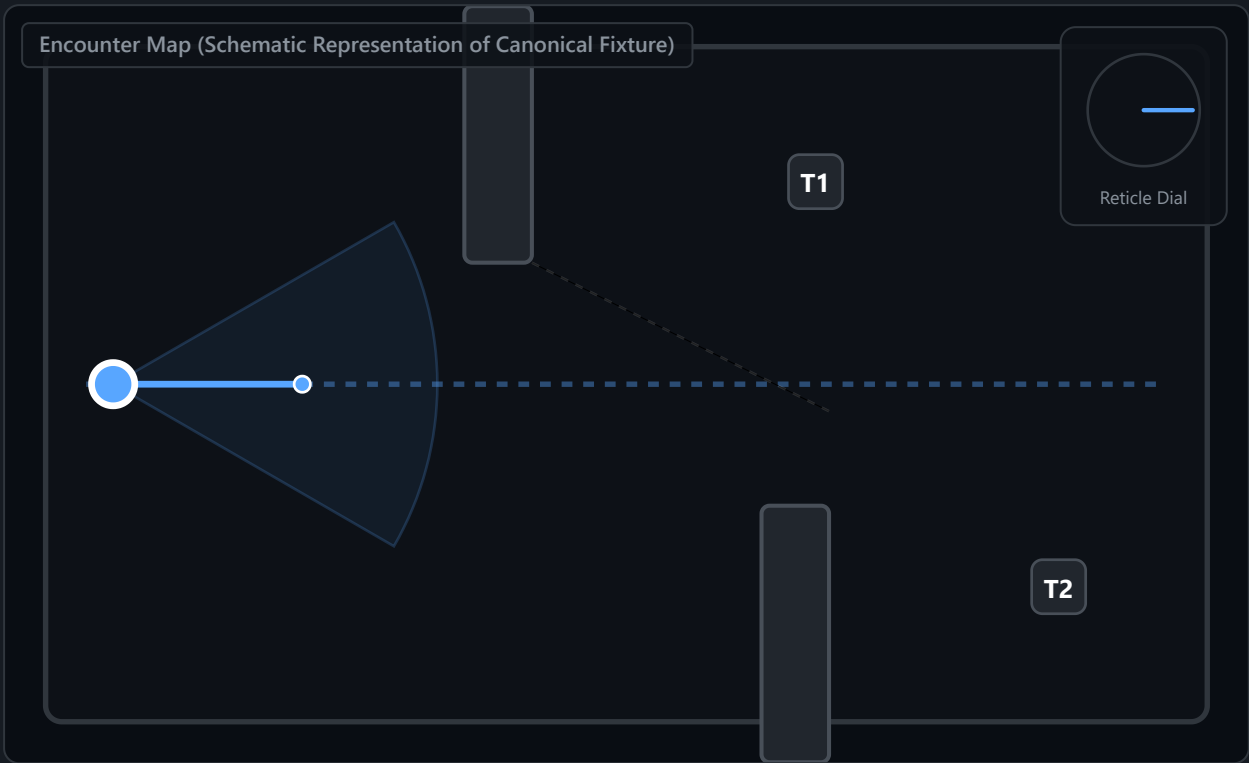


4. The result compiles into a modular contract. Level generators compose local (min, +) matrices without whole-map simulation.

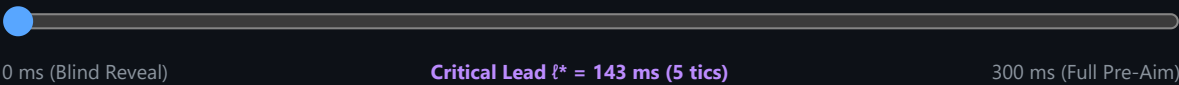
Visual 3: How Much Knowledge Does This Room Require?

Advance knowledge ($a_j(\ell) = \max(0, r_j - \ell)$) allows anticipatory reticle setup before aperture reveal. How many milliseconds of warning makes a lethal room clearable?

Encounter: STIM_06: Double Baffle Pivot ($\ell^* = 143$ ms) ▼



Advance Map Knowledge Lead (ℓ): 0 tics (0.0 ms)



Tactical Margin $M(\ell) = -L^*(\ell)$: -5 tics (-142.9 ms)

-300 ms (Overloaded Ambush)

0 ms (Feasibility Threshold)

+300 ms (Clearable Reserve)

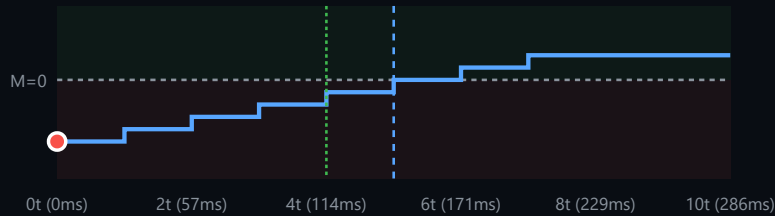


INSUFFICIENT WARNING ($\ell < \ell^* = 5 \text{ tics} / 143 \text{ ms}$)

Reticle cannot pre-align early enough. Non-anticipatory slew creates an unavoidable deadline breach ($L^* = +5 \text{ tics}$).

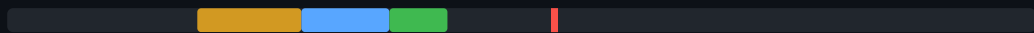
Knowledge Response Function $M(\ell) = -L^*(\ell)$ (Discrete Step Staircase)

— Predicted Lead (ℓ^*_{source}) ... Engine Lead (ℓ^*_{engine})



DYNAMIC SERVICE RECURRENCE ($C_J = \text{MAX}(R_J, \text{MAX}(C_{J-1}, A_J) + Q_{JJ}) + A + P$)

T1 (+64°)



T2 (-62°)



⚡ Snap to Critical Lead (ℓ^*)

▶ Animate Traversal

The Three Canonical Epistemic Classes of Level Geometry

1. Blind-Clearable ($M(0) \geq 0$)

$\ell^* = 0 \text{ ms}$. Geometry permits sequential angle clearance on first sight with zero advance knowledge.

2. Knowledge-Rescuable ($M(0) < 0 \wedge M(\infty) \geq 0$)

$0 < \ell^* < \infty$. Lethal trap to unfamiliar players, but moves across the feasibility boundary with $\geq \ell^*$ warning.

3. Structurally Overloaded ($M(\infty) < 0$)

$\ell^* = \infty$. Flawed geometry. Even with perfect omniscient pre-aim knowledge, the reticle capacity is overwhelmed.

Visual 3B: Decoupling Capacity Gain (ΔM) from Warning Urgency (ℓ^*)

Why the total value of full map knowledge does not tell you how early advance warning must arrive.

STIM_07: Spaced Baffle Pivot

$\ell^* = 114 \text{ ms (4 tics)}$

Initial Deficit: $M_{\text{reveal}} = -4 \text{ tics}$

Pre-Aim Margin: $M_{\text{preaim}} = +3 \text{ tics}$

Total Capacity Gain: $\Delta M = +7 \text{ tics}$

Advance Lead Required:

114.3 ms (4 tics)

STIM_11: Zigzag Flank Pivot

$\ell^* = 171 \text{ ms (6 tics)}$

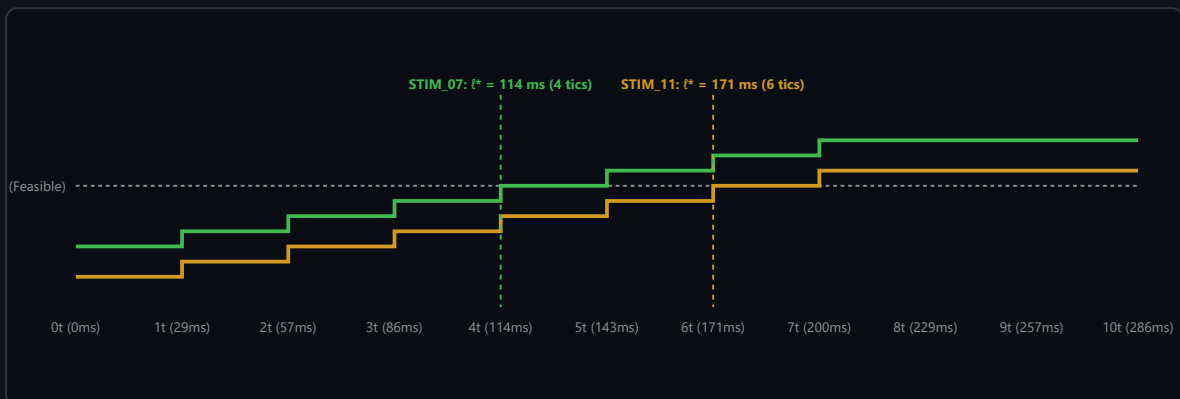
Initial Deficit: $M_{\text{reveal}} = -6 \text{ tics}$

Pre-Aim Margin: $M_{\text{preaim}} = +1 \text{ tic}$

Total Capacity Gain: $\Delta M = +7 \text{ tics}$

Advance Lead Required:

171.4 ms (6 tics) — +50% warning



The Decoupling Principle: Both encounters receive the exact same benefit from full prior knowledge ($\Delta M = +7 \text{ tics (+200 ms)}$). But because STIM_11 started in a deeper deficit (-6 vs -4 tics), it requires **50% more advance actionability** before reaching safety.

LEVEL DESIGN ARCHETYPES

Tactical Readability Atlas: 12 Canonical Spatial Contrasts

Progressive chaptered study of how geometry, timing, traversal trajectory, entry state, map knowledge, and modular boundaries govern FPS clearability.

All Chapters (12)

1. Counting Fails

2. State & Trajectory

3. Map Knowledge

4. Composition & Taxonomy

01

PCG MODULE

Straight Corridor (M01)

EASY BASELINE

CLEARABLE

02

PCG MODULE

Staggered Pie Slicing (M04)

MANY ≠ BAD

CLEARABLE

03

PCG MODULE

Doorway Crossfire (M11)

FEW ≠ SAFE

UNSERVICEABLE

04 ⚡

FROZEN FIXTURE

Route Choice (F05)

PATH MATTERS

DIRECT vs FLANK

05 ⚡

FROZEN SWEEP

Move The Wall (F06)

TIMING MATTERS

EXACT SLIDER

06 ⚡

ARCHETYPE

The T-Junction Baffle

TOPOLOGY ≠ TIMING

FLUSH vs BAFFLED

07 ⚡

ARCHETYPE

Entry Direction

ENTRY STATE MATTERS

WEST vs SOUTH

08

FROZEN STIMULUS

Knowledge Rescue (STIM_07)

INFO MATTERS

RESCUABLE (114ms)

09

FROZEN STIMULUS

Same Gain, Different Need

DECOUPLING

114ms vs 171ms

10

FROZEN STIMULUS

Structurally Overloaded

INFO ISN'T MAGIC

OVERLOADED ($t^* = \infty$)

11 ⚡

PCG CONTRACT

Modular Seam Leakage

COMPOSITION

NAÏVE vs CERTIFIED

12

TAXONOMY

The Three Epistemic Classes

TAXONOMY

3 REGIMES

01 · Straight Corridor (M01)

VISIBLE THREATS

K_static = 1

TACTICAL MARGIN

+28 tics (+800.0 ms)

KNOWLEDGE NEEDED

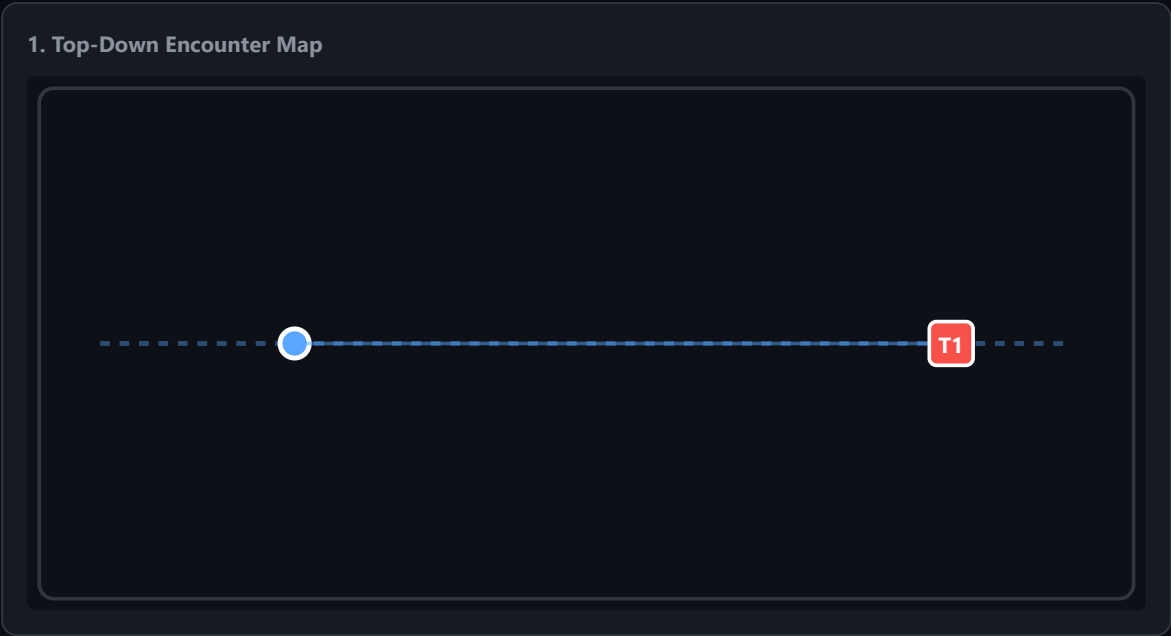
$\varphi^* = 0$ ms

MODEL VERDICT

CLEARABLE

WHAT CHANGED?

Baseline reference: Single forward threat with generous reaction window and zero required angular rotation.



+90°

0°

-90°

0 tics

20 tics

40 tics

60 tics

Procedural Level Linting & Evidence Hierarchy

From local module geometry contracts to global procedural level assembly.

Modular Clearability Certification

Under quiescent boundary ports, global level schedulability reduces to local module verification ($C \equiv D$ theorem):



TACTICAL LINTER VERDICT: Level rejected. Module C contains an unserviceable wide crossfire. Precertified library filtering (Condition E) eliminates 100% of runtime scheduling overhead across 25,000 generated levels.

Scientific Evidence Hierarchy

LAYER	VALIDATION SCOPE	STATUS
1. Formal	Scheduling equivalence & (min, +) dioid associativity	PROVEN
2. Geometry	Sub-millimeter ray-vertex compilation ($K = 8$)	VERIFIED
3. PCG	25,000-candidate sweeps & MAP-Elites non-inferiority	VERIFIED
4. Simulation	60 arenas $\times$ 9,000 35Hz episodes (LOGFO-AUC = 1.0000)	FROZEN
5. ViZDoom	Headless C++ engine translation ($\epsilon_{\text{deploy}} = 3$ tics)	VALIDATED
6. Human	Cognitive familiarity transition (H1-H4)	NEXT

